UXRVT: Unity XR Visualization Toolkit for Large Datasets

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1 Project Goal

This project aims to provide a high-performance, customizable toolkit for authoring basic 2D/3D visualizations in the Unity game engine. The toolkit targets both 2D and immersive environments. Basic interaction techniques such as brushing, linking, and filtering will be supported. We aim to provide a solid framework for building highly customizable basic visualizations that run mainly on the GPU therefore supporting large datasets. The toolkit shall be as agnostic as possible to the input devices and the rendering context. The project can be accessed at any time through the following GitHub repository: https://github.com/walcht/UXRVT.git

2 Used Devices

Since we aim to build a toolkit, there isn't a particular device that we target. Instead, the toolkit should support conventional 2D screen devices (e.g., PCs, tablets, etc.) and XR devices through OpenXR (e.g., magic leap 2).

3 Dataset and Format

There is no particular dataset that we target - instead a showcase of our toolkit will make use of an example of a large, multidimensional dataset. Our toolkit expects structured datasets in CSV, XML, or JSON format (support for additional formats will be easily extensible).

4 Visualization Techniques

4.1 Supported Visualizations

Our toolkit shall support the following basic visualizations:

1. **2D/3D Scatterplots -** with support for custom grid lines, custom axis, orthographic and perspective projections, additional color encoding channel, and highly customizable shapes.

2. **2D/3D Barcharts -** supports the same set of basic functionalities as 2D/3D scatterplots. The shape is customizable and additional color encoding channel is supported.

3. **2D/3D Linecharts -** for 3D linecharts, the 3rd axis is categorical. Supports the same basic set of functionalities as 2D/3D scatterplots.

4. **2D/3D Parallel Coordinates Plot -** the 3D/2D parallel coordinates plot supports the same set of basic functionalities supported by 2D/3D scatterplots. The 3D version simply uses a plane rather than a 1D axis.

5. **3D Plots -** this can be thought of as a 3D linechart with a numerical 3rd axis.

Scatterplot matrix plot will optionally be supported depending on time constraints. However, we believe that this plot can already be easily authored using multiple 2D/3D scatterplots.

4.2 Supported Interactions

All the supported visualizations mentioned above support the following interaction techniques:

1. **Brushing** - selection of a subset of the data points with custom visual feedback (e.g., highlight color).
2. **Filtering** - limit the amount of displayed data points through selection. Advanced filtering techniques will not be supported.
3. **Linking** - show how a set of data points in a particular plot is related to the same set in a different plot. The visualization feedback that shows the linking between different plots is customizable (e.g., by default it is a line but can be changed/disabled).
4. **Details on Demand** - when hovering over a datapoint, show a customizable tooltip for additional details.
5. **Transform Operations** - scalling, rotating, and translating (i.e., moving around) any visualization object is supported.

5 Mock-up

A mock-up of the end result of our visualization toolkit is shown in 1. The mock-up (and this whole project) is largely influenced by the existing immersive analytics toolkit IATK. Even though we largely share the same goal, our toolkit aims to be more customizable, and more interactive. Our toolkit also focuses more on providing support for both: conventional 2D screens and XR devices.

6 Main Workstation Hardware

The main hardware on which implementations will be done is the following: CPU: AMD Ryzen 5 7600X 6x 4.70GHz, GPU: NITRO+ AMD Radeon RX 7800 XT 16G GDDR6, RAM: 32GBs 5600Mhz DDR5 RAM, and SSD: Kingston NV2 NVMe PCIe 4.0 Internal SSD 1TB.

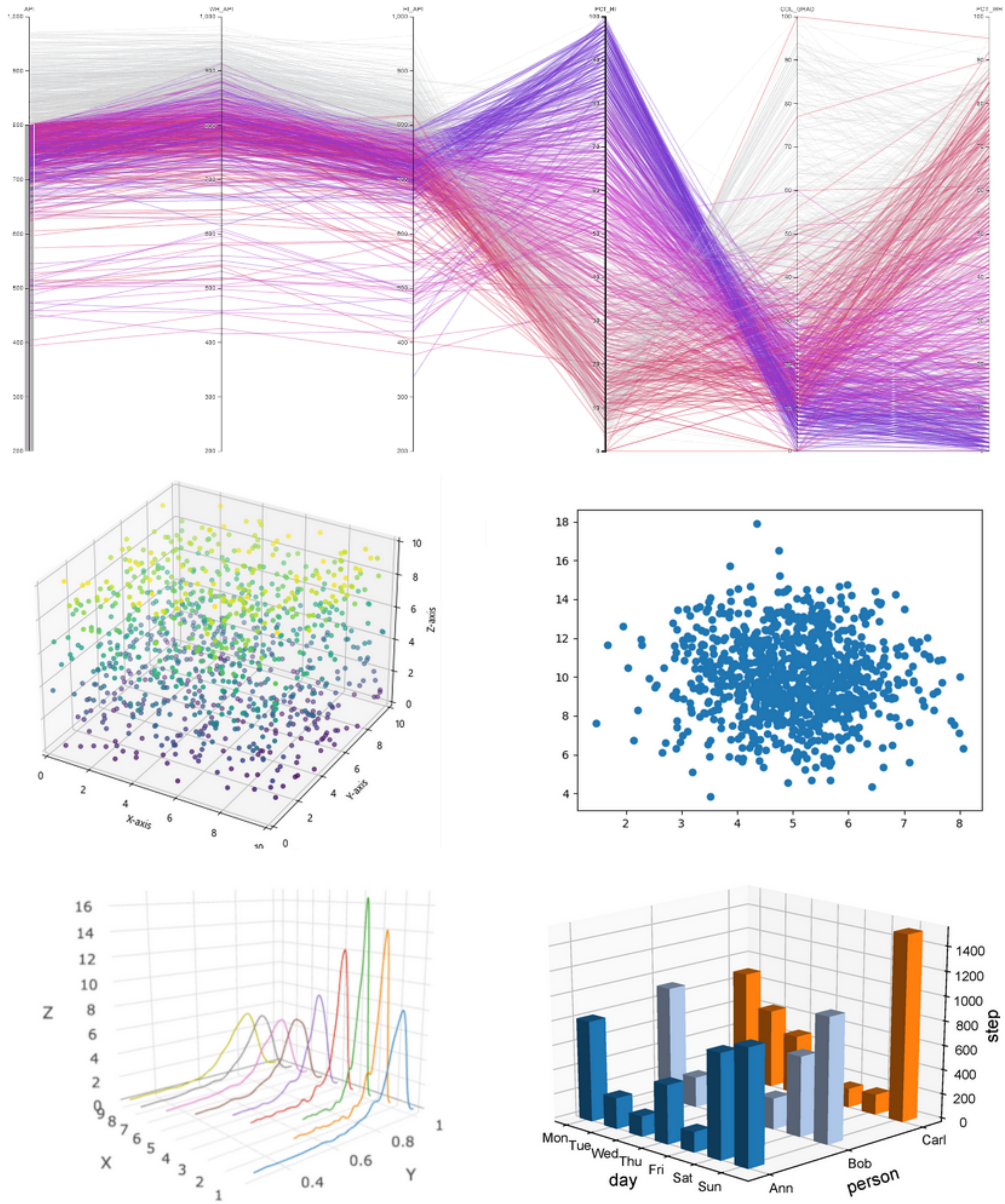

Figure 1: UXRVT supported visualizations